

BASICS OF NETWORKS & NETWORK TOPOLOGIES

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Topics to be covered

Networks

- i. **Three network criteria**
- ii. **Network physical structure**
 - a. **Type of connection**
 - b. **Physical topology**
 - 1. **Mesh**
 - 2. **Star**
 - 3. **Bus**
 - 4. **Ring**

NETWORKS

- A **set of devices** (or *nodes*) connected by **communication links**.
- A **node** can be a computer, printer, or any other device capable of sending data and/or receiving data generated by other nodes on the network.
- A **link** is a communications pathway that transfers data from one device to another.

Three Network Criteria

1. Performance
2. Reliability
3. Security

Three Network Criteria

1. Performance

- 2 important ways are to measure performance :
 1. **Transit time** is the amount of time required for a message to travel from one device to another.
 2. **Response time** is the elapsed time between an inquiry and a response.
- The performance of a network depends on factors such as –
 - the **number of users**,
 - the **type of transmission medium**,
 - the **capabilities of the connected hardware**, and
 - the **efficiency of the software**.
- 2 **Networking Metrics** to **evaluate** performance are **throughput and delay**.
- We often need more throughput and less delay. However, these two criteria are often contradictory. If we try to send more data to the network, throughput increase but the delay increase too because of traffic congestion in the network.

Three Network Criteria

2. Reliability

- In addition to accuracy of delivery, network reliability is measured by :
 - the **frequency** of failure
 - the **time it takes for a link to recover from a failure**, and
 - the network's **robustness in a catastrophe**.

Three Network Criteria

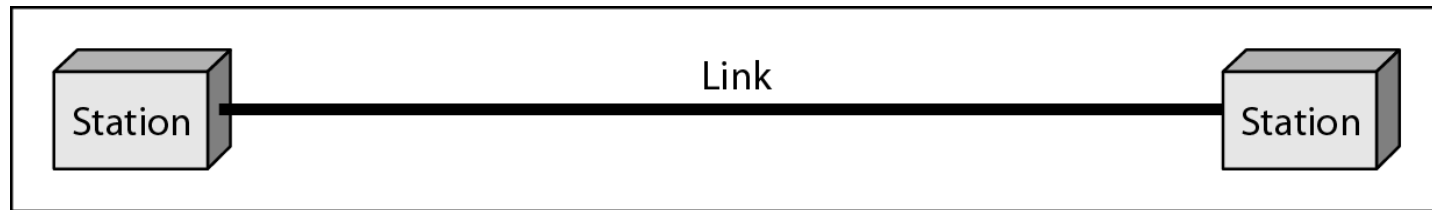
3. Security

- Network security issues include :
 - **protecting** data from **unauthorized access**,
 - **protecting** data from **damage and development**, and
 - **implementing policies and procedures** for recovery from breaches and data losses.

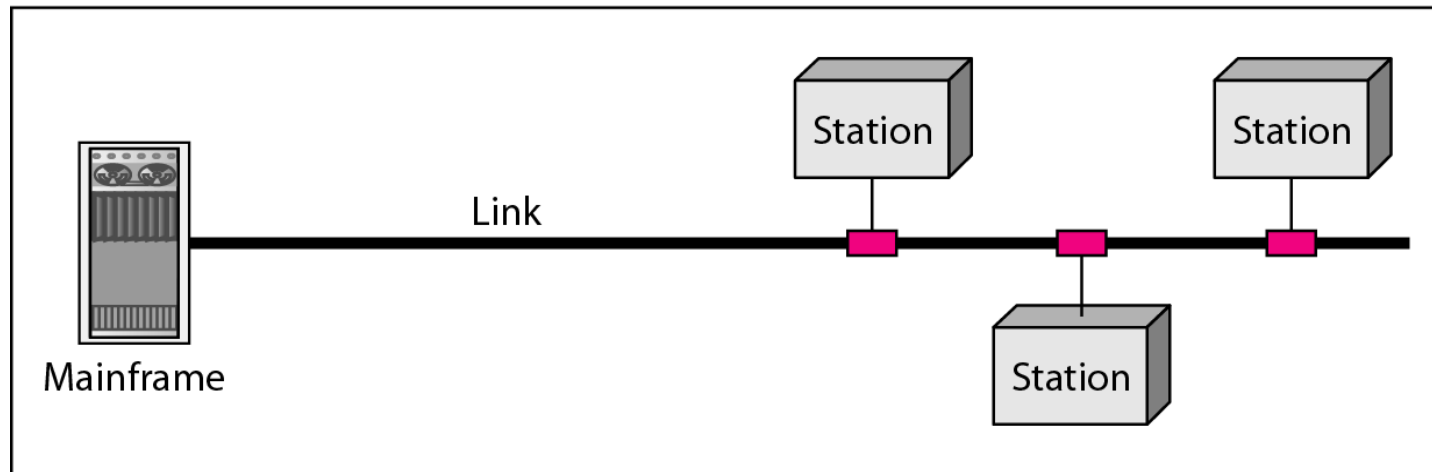
Physical Structures - Type of Connection

- **Point-to-Point** : a dedicated link between two devices.
 - Link options : wire or cable to connect the two ends or microwave or satellite links.
- **Multipoint** (also called **multidrop**): more than two specific devices share a single link.
 - Here the **capacity of the channel is shared**, either spatially or temporally.
 - If several devices can use the link simultaneously, it is a ***spatially shared*** connection.
 - If users must take turns, it is a ***timeshared connection***.

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point

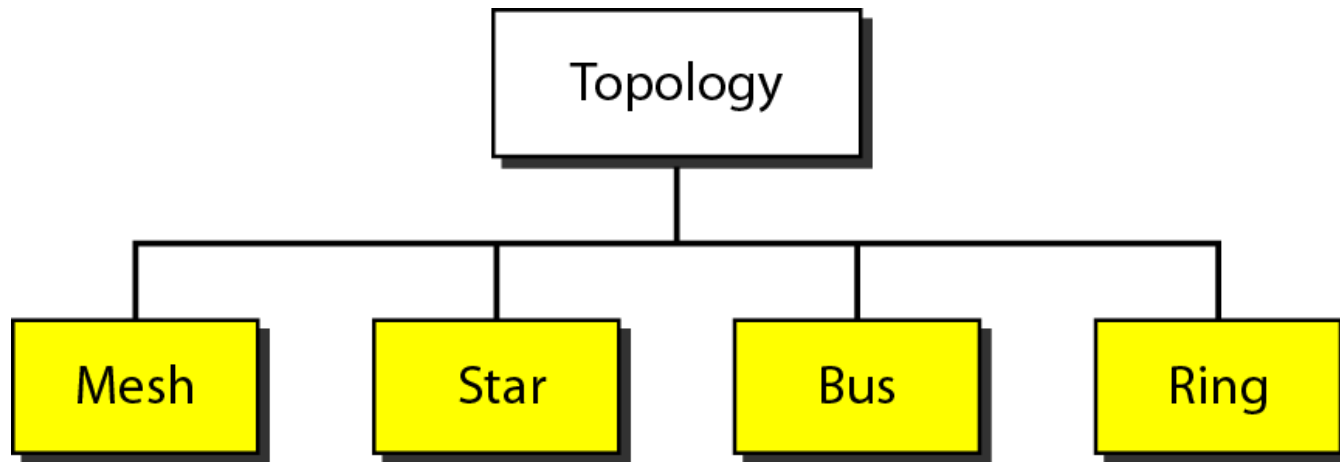


b. Multipoint

Physical Structures – **Physical Topology**

- Refers to the way in which a **network is laid out physically** OR the **geometric representation** of the relationship of all the links and linking devices (usually called nodes) to one another.
- Four basic topologies possible:
 1. mesh
 2. star
 3. bus
 4. ring

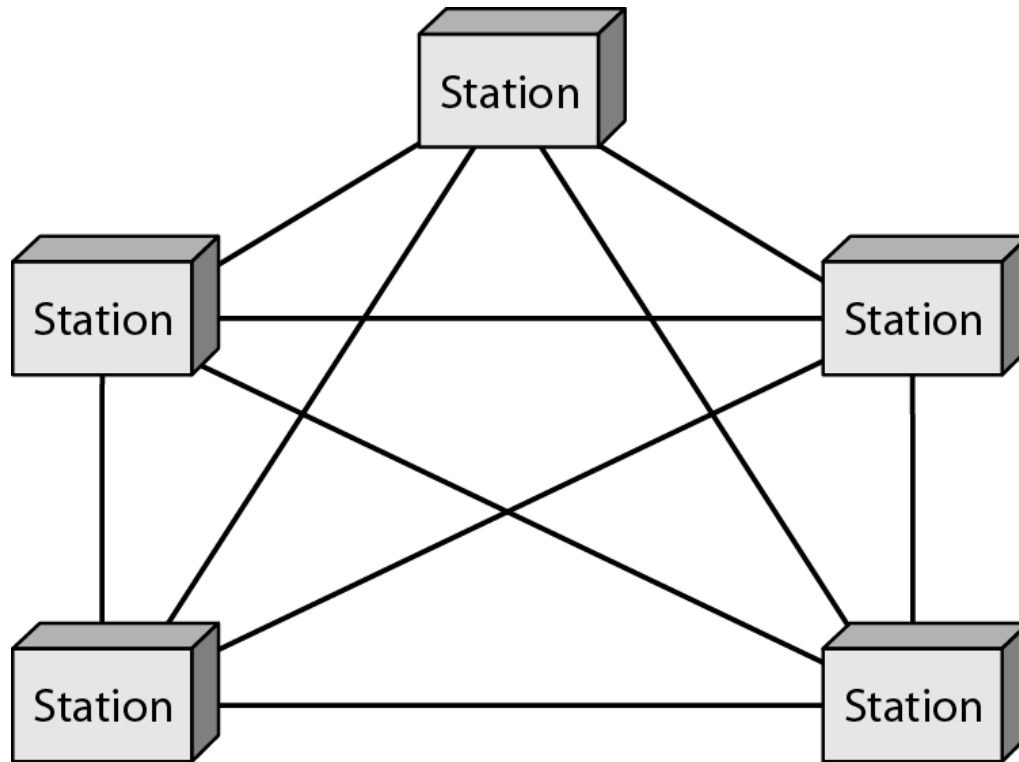
Figure 1.4 *Categories of topology*



Mesh topology

- Every device has a **dedicated point-to-point link** to every other device.
- In a fully connected mesh network with *n* nodes, we need _____ ***simplex physical links***.
- If each physical link allows duplex mode, we need _____ **duplex-mode links**.
- i/e., every device on the network must have _____ ***I/O ports*** (see Figure 1.5) to be connected to the other $(n - 1)$ stations.

Figure 1.5 *A fully connected mesh topology (five devices)*



Advantages & disadvantages of Mesh Topology

➤ Advantages

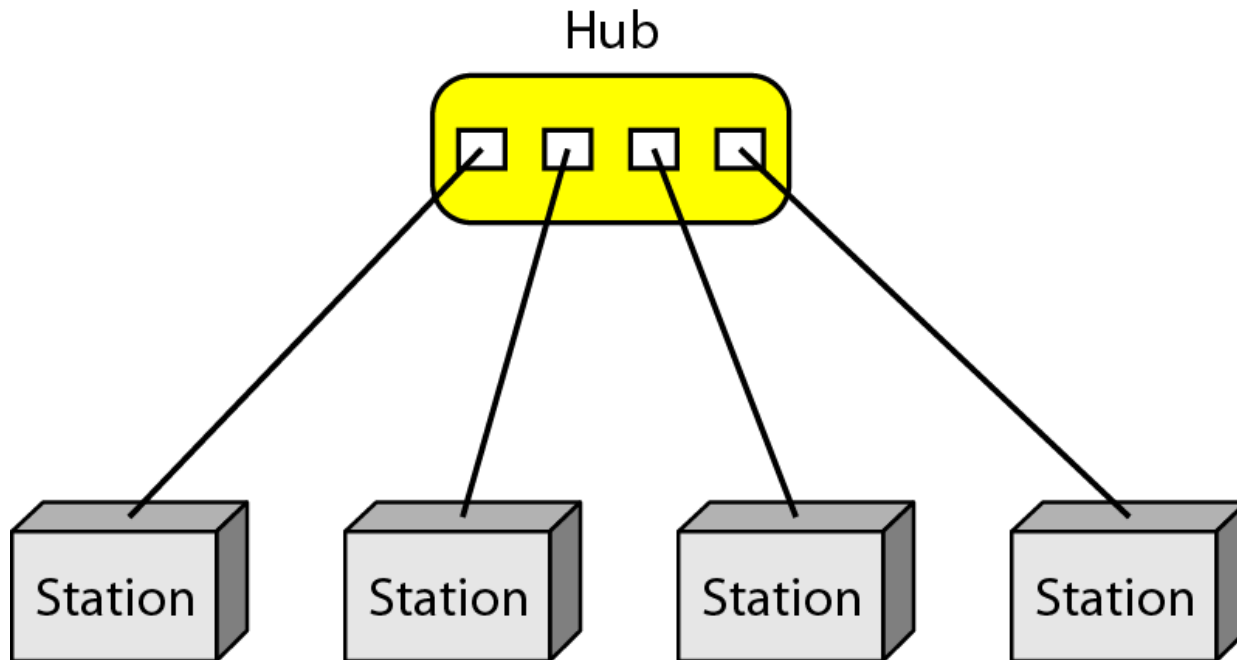
- Dedicated links eliminate the traffic problems
- **Robust**: one unusable link does not incapacitate the entire system
- **Privacy or security**
- Easy **fault identification & fault Isolation** : Traffic can be routed to avoid links with suspected problems

➤ Disadvantages

- Large number I/O ports & **cabling** required
- Installation & reconnection are **difficult**
- Wiring can be greater than **available space**
- Hardware required to connect each link (I/O ports & cable) can be **very expensive**

Example: the connection of **telephone regional offices** in which each regional office needs to be connected to every other regional office.

Figure 1.6 *A star topology connecting four stations*



Star Topology

- Each device has a dedicated point-to-point link **only to a central controller** (Ex: **hub**)
- No direct traffic between devices. The controller acts as an **exchange**.

Advantages & disadvantages of Star Topology

➤ Advantages

- **Less expensive** than mesh topology
- **Installation & reconfiguration** are **easy**
- Required **less cabling** than mesh topology
- **Robust**: If one link fails, only that link is affected
- **Easy fault identification & fault Isolation**

➤ Disadvantages

- **Dependency** of whole topology **on single point**

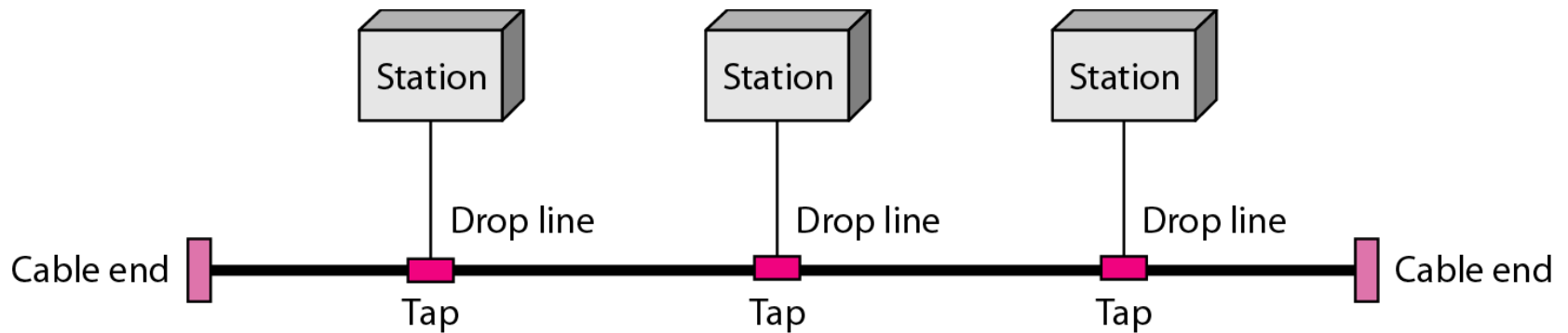
Application: High Speed LAN

Example: used in local-area networks (LANs)

Bus Topology

- One **long cable** acts as a **backbone** to link all the devices in a network (see Figure 1.7)
- Nodes are connected to the bus cable by **drop lines** and **taps**.
- As a signal travels along the backbone, some of its energy is transformed into heat making it weaker. For this reason there is a **limit on the number of taps** and on **the distance between those taps**.

Figure 1.7 *A bus topology connecting three stations*



Advantages & disadvantages of Bus Topology

➤ Advantages

- **Ease of installation**
- Requires **less cabling**: Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths

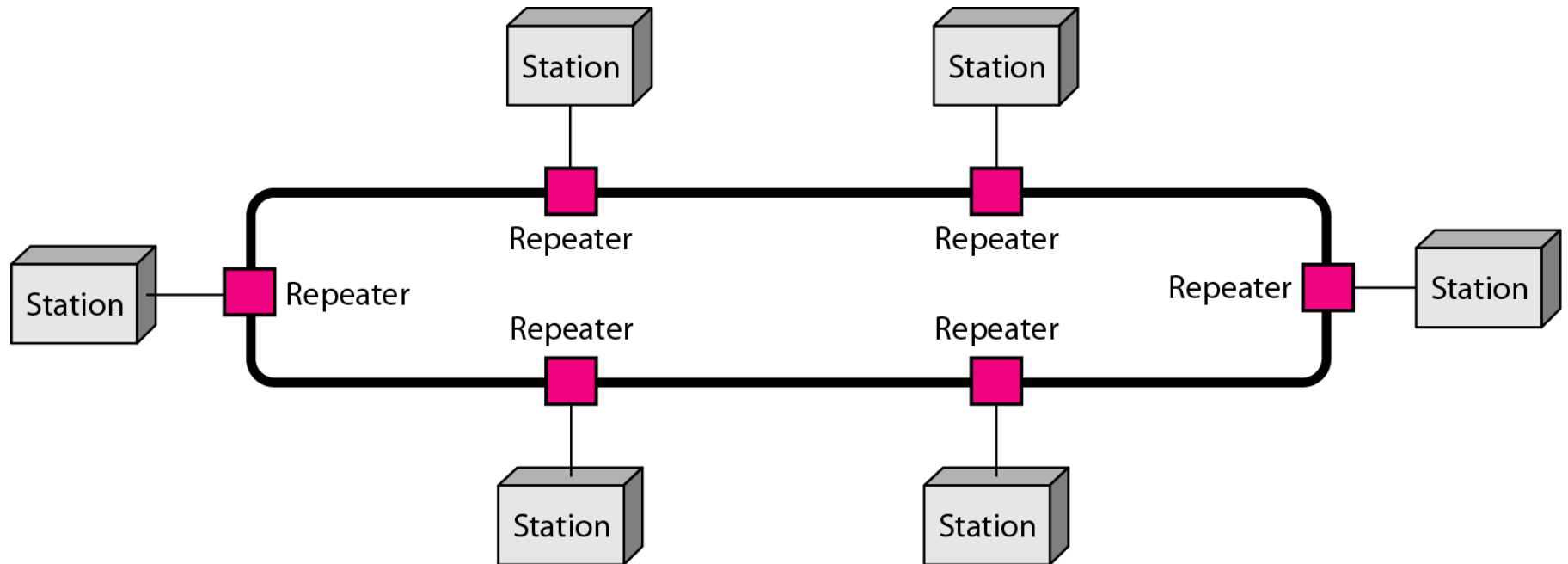
➤ Disadvantages

- **Difficult reconnection & fault isolation**
 - bus is usually designed to be optimally efficient at installation. Hence, difficult to add new devices
- **Signal reflection at the taps** can cause degradation in quality
- A fault or break in the bus cable **stops all transmission**

Ring Topology

- Each device has a dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination
- Each device incorporates a **repeater**: repeater regenerates the bits and passes them along when a device receives a signal intended for another device

Figure 1.8 *A ring topology connecting six stations*



Advantages & disadvantages of Ring Topology

➤ Advantages

- Easy to install & reconfigure
- fault isolation is simplified

➤ Disadvantages

- **Unidirectional traffic** : a break in ring (such as a disabled station) can disable the entire network.

Hybrid Topology

- Combining different topologies

Hybrid Topology

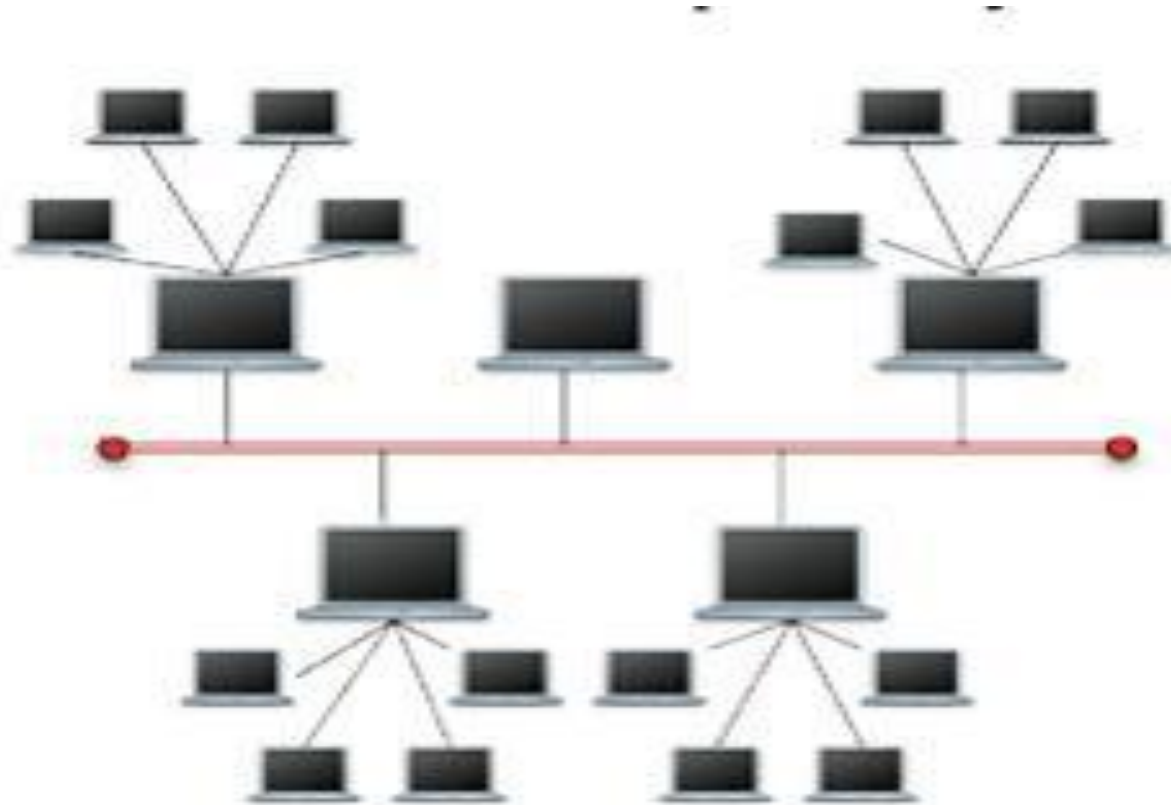
- Nowadays Hybrid topology is used
- **Hybrid Topology**: It is a mix of two or more topologies.
- **Example**:
 - There are two networks; one is built from the star topology and another is built from the bus topology. If we connect both networks to build a single large network, the topology of the new network will be known as the hybrid topology.
- We can combine any topology with another topology.
- This combination of topologies is done according to the requirements of the organization
- In modern network implementations, the hybrid topology is mostly used to mix the wired network with the wireless network.
- **Example**: Star-Bus , Star-Ring

Hybrid Topology

- **Hybrid topology**
 - **Star-Bus:** The star bus topology is combined with the linear bus in large networks. In such cases the linear bus provides a backbone that connects multiple stars.
 - **Star-Ring:** The Star-Ring is a combination of ring and star topologies. The hubs in a star ring are connected in a star pattern by the main hub.

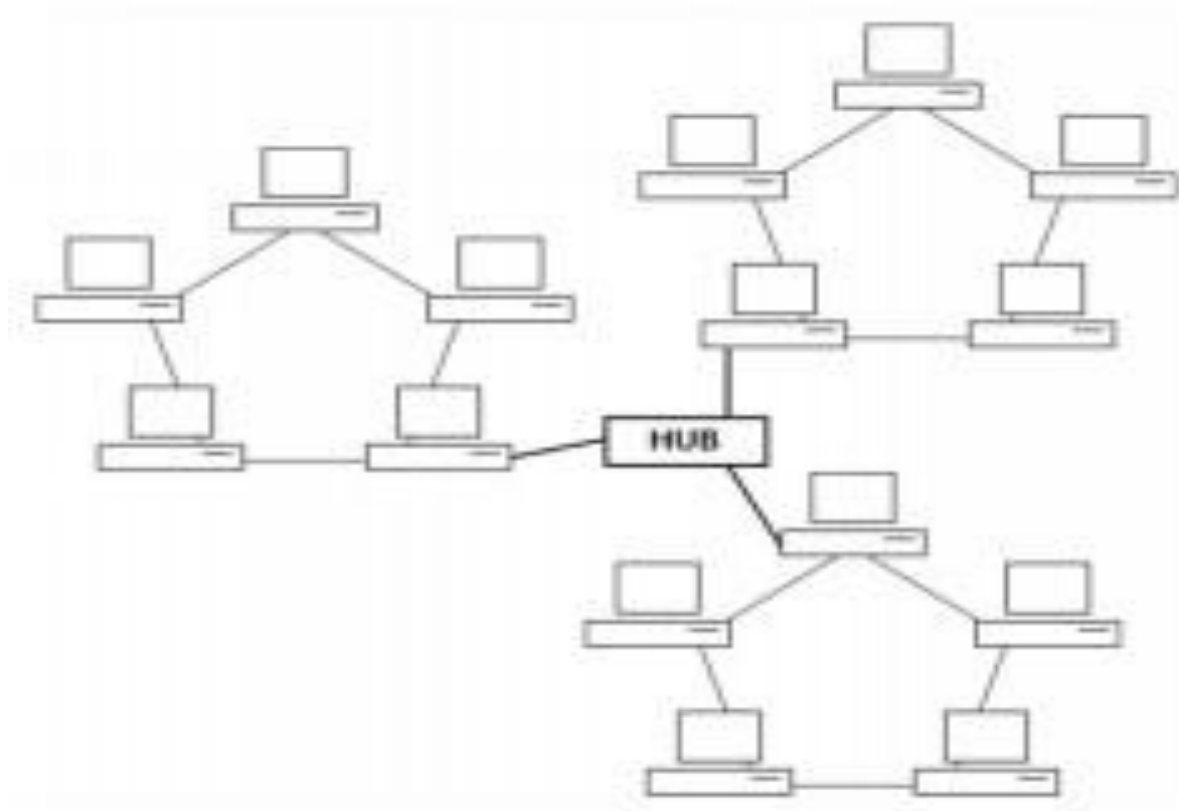
Hybrid Topology

- Hybrid topology-Example: Star-Bus



Hybrid Topology

- Hybrid topology-Example: Star-Ring



Hybrid Topology **Advantages**

Reliable: Fault detection and troubleshooting is easy. The part in which fault is detected can be isolated from the rest of the network and required corrective measures can be taken without affecting the functioning of the rest of the network.

- **Scalable**: It is easy to increase the size of network by adding new components without disturbing existing architecture.
- **Flexible**: It can be designed according to the requirements of the organization and by optimizing the available resources. Special care can be given to nodes where traffic is high as well as where chances of fault are high.
- **Effective**: It can be designed in such a way so that strengths of constituent topologies are maximized while their weaknesses are neutralized.

Hybrid Topology **disadvantages**

- **Complexity of Design**: It is not easy to design this type of architecture. Configuration and installation process needs to be very efficient
- **Costly Hub**: The hubs used to connect two distinct networks are very expensive because they are different from the normal ones. They are intelligent enough to work with different architectures and should function even if one part of the network is down
- **Costly Infrastructure**: Hybrid infrastructure are usually large in scale and hence they require a lot of cables, cooling systems and sophisticated network devices

